A Dynamical Systems Framework for Adaptive Memory: Four Scaling Laws of Plastic Information Substrates

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Abstract

We present a unified framework for understanding adaptive information systems as physical dynamical systems. Rather than treating representations as fixed parameters in optimization, we model them as entities in a dissipative field governed by birth, death, diffusion, and selection. We show that four fundamental constraints—separation, turnover, propagation, and stability—jointly determine the operating regime of any plastic information substrate. These constraints are empirically validated in a viability-constrained coupled map lattice (VCML) and cross-validated on real language-model embeddings (sentence-transformers); their extension to other substrates remains a hypothesis, not yet a result. We derive scaling laws for capacity, bandwidth, coherence, and persistence as consequences of these constraints. More importantly, we show that three of these laws interact: their intersection produces a fifth derived quantity, the *coherence efficiency* $\eta_c \approx (L/D)^2$, which measures what fraction of theoretical capacity is coherently accessible and imposes a matching condition $C\sqrt{\kappa/\nu} \approx D$ coupling propagation length, zone scale, and system size. This coupling transforms four independent descriptions into a single design problem: scaling one control parameter without adjusting the others degrades coherent capacity as $\eta_c \sim N^{-1}$. We additionally construct a *conceptual* phase diagram mapping transformers, Hopfield networks, and biological memory systems onto the same parameter space; these placements are theoretical inferences, not empirical measurements of those systems.

1 Introduction

Modern machine learning lacks a *physical theory* of representations. Instead, we have:

- **Optimization theory**: representations are parameters updated via gradient descent.
- **Information theory**: representations are codewords minimizing mutual information.
- **Neuroscience**: representations are neural firing patterns.

But none of these frameworks explains why representations in *adaptive* systems must obey scaling laws, why they decay at a particular rate, or why stability and plasticity coexist rather than trade off.

The prerequisite condition. Before any law of adaptive memory can hold, a more fundamental condition must be satisfied. Paper 0 [1] establishes it by ablation: disabling the copy-forward loop (the intergenerational transmission pathway) collapses spatial structure to exactly zero, regardless of input, initialization, or turnover rate. The system becomes a transient reservoir—information survives for at most one generation. This is the *Turnover–Transmission Principle*:

> *Persistent information in systems with mandatory component turnover requires an intergenerational transmission pathway. Without it, the system is a transient reservoir: memory lifetime is bounded by individual lifespan.*

This principle is substrate-independent. It holds for DNA and epigenetic inheritance (remove it and evolution collapses), synaptic consolidation (remove it and learning collapses), and distributed state handoff (remove it and collective memory disappears). Paper 0 demonstrates it in its minimal computational form. The four laws below characterize the *limits* on what that transmission can achieve—but only once the prerequisite is satisfied.

Papers 1–4 [2, 3, 4, 5] discovered four empirical laws:

1. **Capacity law** [2]: separable representations scale as $N \propto \tau^{-d_{\text{seen}}}$.
2. **Bandwidth law** [3]: sustainable turnover scales as $B \approx N_{\max}/\tau_{1/2}$.
3. **Propagation law** [4]: coherence distance scales as $L \propto \sqrt{\kappa/\nu}$.

4. **Stability constraint** [5]: optimal gating strength tracks the noise level of the environment.

This paper shows that all four laws are *necessary consequences* of treating representations as particles in a dissipative field. This perspective transforms an empirical collection into a *unified theory*.

The central new result is that these laws do not operate independently. Section 5 derives a fifth quantity from the interaction of Laws 1, 2, and 3—the *coherence efficiency* $\eta_c \approx (L/D)^2$—and shows that it imposes a matching condition $C\sqrt{\kappa/\nu} \approx D$ linking propagation length, zone width, and system size. This matching condition is the first place where the laws *interact* rather than merely coexist. It is where the theory becomes a *system*.

2 Plastic Information Substrates

2.1 Definition

A *plastic information substrate* is a dynamical system where:

1. **Representations** are discrete entities (sites, nodes, or cells) indexed by location and state.

2. **Birth**: new representations are generated from environmental input or seeded from neighbors.

3. **Death**: representations are eliminated stochastically or when viability thresholds are crossed.

4. **Learning**: representations are modified locally via contrastive or Hebbian updates.

5. **Coupling**: information propagates between neighboring representations via diffusion or relay.

6. **Filtering**: updates to long-term memory are gated by a persistence threshold, suppressing noise.

This definition encompasses:

- Coupled map lattices with viability constraints (our VCML).

- Dynamical systems models of hippocampal place cells.

- Associative memory networks with pruning and learning rules.

- Biological neural tissue with neurogenesis and apoptosis.

It *excludes*:

- Static parameter spaces (transformers, feedforward networks).
- Systems without explicit death or turnover (standard neural networks).

2.2 Core Properties

Plastic substrates exhibit five hallmark properties:

1. Mandatory turnover Every representation has a finite lifetime. Stability emerges from the *structure* written to long-term memory, not from persistent individual states.

2. Local coupling Information spreads through nearest-neighbor interactions. Global patterns emerge without central control.

3. Contrastive learning Updates are driven by perturbations: midterm memory accumulates deviations from baseline; consolidation filters surprise.

4. Viability gating Birth, learning, and death are controlled by thresholds (age, consolidation streak, energy cost).

5. Substrate independence (partial) The capacity law holds across lattices (VCML-L, Papers 1–4) and real embedding spaces (VCML-E, Paper 1 Exp. 8); graph-coupled substrates (VCML-G) are architecturally defined but not yet fully validated empirically. Substrate independence is an open hypothesis supported by two of three cases. The present results demonstrate the laws in VCML substrates and partially in embedding spaces; extending them to other substrates remains a hypothesis rather than a demonstrated universality.

3 The Four Constraints

Notation and Scale Conventions

Unless otherwise stated, N, B, and L are defined at the same substrate scale: N denotes the number of active representational slots in the region under consideration, B denotes the replacement rate of those same slots per unit time, and L denotes the coherence length measured within that same region. When the framework is applied hierarchically (e.g., geographic relay), each region obeys the laws locally, and inter-regional coupling introduces a second scale rather than redefining the original variables.

We use three distinct symbols for different control timescales: τ (geometric separation scale, Paper 1), τ_{hl} (slot lifetime controlling bandwidth, Paper 2), and τ_g (consolidation gate threshold, Paper 4).

The geometric dimension d_{seen} in the capacity law refers to the intrinsic dimensionality of the representational manifold; L in the propagation law refers to the physical coherence length in substrate space. These quantities interact through the coherence efficiency η_c (Section 5) but are not the same variable.

3.1 Constraint 1: Separation (Paper 1 — Capacity)

For representations to be distinguishable, their states must span sufficient dimensionality.

Mechanism: Fisher separation between zones. A zone is separable from its neighbors only if the representation space $\mathbb{R}^{H_S}$ is large enough to encode their locations.

Law:

$$N_{\max} \propto \tau^{-d_{\mathrm{seen}}} \tag{1}$$

where d_{seen} is the effective dimensionality visible to the separator, and τ is the learning timescale.

Physical meaning: Fine-grained separation demands finer oscillations and shorter consolidation times. Capacity scales inversely with learning speed.

3.2 Constraint 2: Turnover (Paper 2 — Bandwidth)

Representations that live forever cannot learn. But high turnover rates guarantee structure is destroyed.

Mechanism: Birth–death equilibrium. The long-term memory capacity ($N_{\max} \cdot H_S$ dimensions) must be refreshed faster than the environment changes, but slow enough that structure persists through several generations.

Law:

$$B_{\text{sustainable}} \approx \frac{N_{\max}}{\tau_{1/2}} \tag{2}$$

where $\tau_{1/2}$ is the half-life of a representation and B is the throughput (bits per unit time).

Physical meaning: Bandwidth is fundamentally limited by the rate at which you can replace and re-encode representations. It is not a network property (width, depth, parameters), but a *dynamical* property (turnover rate).

3.3 Constraint 3: Propagation (Paper 3 — Coherence Distance)

For distributed representations to form unified patterns, information must propagate faster than it decays.

Mechanism: Diffusion–decay competition. A perturbation diffuses through coupling strength κ but decays due to turnover ν and field decay.

Law:

$$L \propto \sqrt{\frac{\kappa}{\nu}} \tag{3}$$

where L is the coherence distance (spatial scale over which zones remain entrained), κ is the coupling strength, and ν is the effective decay rate.

Physical meaning: No amount of coupling can overcome intrinsic decay. The coherence distance is set by the balance between spreading and forgetting. Larger systems (more zones) require stronger coupling or slower decay.

3.4 Constraint 4: Stability (Paper 4 — Persistence Filtering)

Plastic systems must distinguish signal from noise without rigidifying.

Mechanism: Surprise–persistence gating. Consolidation to long-term memory occurs only after a representation has been consistently selected across multiple timescales.

Law:

$$\tau^*_{\text{gate}} \propto \eta_{\text{noise}} \tag{4}$$

where η_{noise} is the fraction of inputs that are corrupted. The gate is not a universal critical point: in clean environments, lower τ_g maximizes structure (more writes = more zone differentiation); in noisy environments, higher τ_g is essential to filter transient perturbations. At 25% noise, a gated system ($\tau_g = 20$) outperforms an ungated one ($\tau_g = 1$) by 2.52× in spatial structure.

Physical meaning: The gate is a noise filter, not a phase-transition dial. Optimal gating tracks the signal-to-noise ratio of the environment. This resolves the stability-plasticity dilemma: there is no universal critical point; the threshold adapts to the noise regime. In biological systems, neuromodulatory signals encoding environmental uncertainty could implement this adaptation.

4 Unified Field Description

4.1 The Conceptual Equation

Integrating the four constraints yields a single conceptual evolution equation:

$$\frac{\partial M}{\partial t} = \kappa \nabla^2 M - \nu M + F_{\text{sep}}(M, I) + S(M) \tag{5}$$

where:

- M is the representation field (long-term memory).
- $\kappa \nabla^2 M$ is spatial diffusion (Constraint 3: propagation).
- $-\nu M$ is exponential decay due to turnover (Constraint 2: bandwidth).
- $F_{\text{sep}}(M, I)$ is routing/separation driven by input I (Constraint 1: capacity).
- $S(M)$ is the stability filter, a nonlinear gating function (Constraint 4: stability).

4.2 This is Not a Derivation

This equation is *not* derived from first principles. Rather, it is the *continuum abstraction* of the four-step VCSM algorithm. It summarizes the qualitative structure: spreading, decay, input-driven updates, and filtering.

The equation should therefore be interpreted as a phenomenological model capturing the dominant dynamical terms of the VCSM algorithm rather than a microscopic derivation.

It is useful because it:

1. Shows why the four constraints are coupled (each term affects long-term dynamics).
2. Connects plastic substrates to classical PDEs in physics (reaction–diffusion, Ginzburg–Landau).
3. Suggests new substrates by varying κ, ν, F_{sep}, or S.

5 Coherence Efficiency: The Central Synthesis Result

The four constraints of Section 3 describe limits in isolation: capacity bounds how many slots fit, bandwidth bounds how fast they turn over, propagation bounds how far information travels, and stability bounds noise tolerance. Considered separately, they are four *parallel* constraints. The central synthesis of this paper is that Laws 1–3 do not merely coexist—they *interact*, producing a derived quantity that turns the framework into a coupled design problem.

5.1 Definition

Let D be the representational zone width (the spatial scale over which the system is expected to form a unified structure) and $L \approx C\sqrt{\kappa/\nu}$ be the correlation length from Constraint 3. The *coherence efficiency* is:

$$\eta_c \approx \left(\frac{L}{D}\right)^2 \tag{6}$$

η_c answers a question that the four earlier laws cannot individually answer: *how much of theoretical capacity is actually usable?* When $L \ll D$, structure forms in patches smaller than a zone and $\eta_c \ll 1$: the system holds high nominal capacity ($N_{\max}$ slots) but only a fraction is coherently organised. When $L \approx D$, the copy-forward loop spans the full zone and $\eta_c \approx 1$. This is not a new free parameter or a new empirical law—it is a *derived constraint* from the interaction of Laws 1, 2, and 3.

5.2 Empirical Support

V108 (Paper 3) varies zone width D while holding coupling fixed. Doubling D halves efficiency; quadrupling D reduces it fivefold. The observed scale ratio (0.19 at 4× scale) is consistent with $\eta_c \propto D^{-2}$ (predicted 0.25), confirming the square scaling to within measurement noise.

5.3 The Matching Condition

An efficient system requires $L \approx D$:

$$C\sqrt{\frac{\kappa}{\nu}} \approx D \quad \Longrightarrow \quad \frac{\kappa}{\nu} \approx \frac{D^2}{C^2} \tag{7}$$

Since zone width $D \propto N_{\text{active}}^{1/2}/N_{\text{zones}}$ for a square region, the required coupling-to-noise ratio must scale with system size:

$$\frac{\kappa}{\nu} \propto \frac{N_{\text{active}}}{N_{\text{zones}}^2} \tag{8}$$

This is not a new free parameter. It is a *compatibility constraint* between the three control parameters (τ, τ_{hl}, κ/ν) and system size N. A system that satisfies the capacity law but ignores the matching condition achieves only fraction η_c of its theoretical capacity. The V93 finding—that optimal wave ratio (controlling ν) scales linearly with N_{active}—is the empirical counterpart: coupling must grow with system size to maintain $L \approx D$.

5.4 Design Implication

Scaling a plastic substrate while holding coupling fixed degrades coherence efficiency as $\eta_c \sim N^{-1}$. Geographic relay (Papers 3, V100–V102) is the architectural response: operate each region at scale N_r satisfying $L \approx D_r$, then link regions hierarchically. This recovers $\eta_c \approx 1$ at any total system size, at the cost of an inter-region coupling mechanism.

The matching condition (7) is where Laws 1–3 first interact. It motivates the phase diagram in the following section: the adaptive window $\nu \approx \tau_g^{-1}$ is the temporal analogue—a matching condition between turnover rate and gating threshold that parallels the spatial matching condition between coupling and zone scale.

6 The Feasible Regime: Joint Constraints

6.1 Why the Three Laws Cannot Be Independently Maximized

The four laws of Section 3 appear to describe independent axes: capacity is controlled by τ, bandwidth by $\tau_{1/2}$, and propagation by κ/ν. A designer might expect to maximize all three simultaneously by choosing each parameter freely. This expectation is wrong. The three control parameters are linked through the system size N_{active} and zone width D, and their joint constraints define a *feasible regime*—a region of parameter space in which the substrate can simultaneously maintain representations, refresh them at the required rate, and propagate structure across each zone.

6.2 Three Failure Modes

Violating any one of the three constraints produces a qualitatively distinct failure:

Failure mode	**Cause**	**Signature**	**From l**
Capacity failure	τ too large, zones collide	sg4 $\to 0$; zones merge	Paper 1
Bandwidth failure	$\tau_{1/2}$ too long, drift exceeds refresh	structure lags environment	Paper 2
Propagation failure	$L \ll D$, copy-forward cannot span zone	patchy structure; $\eta_c \ll 1$	Paper 3

The three failure modes are independent: a system can satisfy any two while violating the third. Capacity can be adequate (τ small enough) while propagation fails ($\kappa/\nu \ll D^2/C^2$); bandwidth can be sufficient ($\tau_{1/2}$ short) while capacity fails (τ too large). The feasible regime is the intersection of all three satisfied simultaneously.

6.3 The Feasible Regime

Combining the three constraints, a plastic substrate maintains stable spatial structure only when:

$$\underbrace{\tau^{-d} \geq N_{\text{zones}} \cdot r_{\min}}_{\text{capacity constraint}} \quad \cap \quad \underbrace{\frac{N_{\max}}{\tau_{1/2}} \geq B_{\text{env}}}_{\text{bandwidth constraint}} \quad \cap \quad \underbrace{C\sqrt{\frac{\kappa}{\nu}} \geq D}_{\text{propagation constraint}} \tag{9}$$

where $r_{\min}$ is the minimum per-zone representation resolution required, B_{env} is the rate at which the environment changes, and D is the zone width. Together these form a *feasibility polytope* in the space of (N, B, L) (Figure 1): the achievable region is not the full positive orthant but a bounded subvolume.

6.4 System-Size Dependence and the Optimal Scale

The feasibility constraints interact through system size N_{active}. Increasing N_{active} while holding coupling κ/ν fixed:

1. Enlarges zone width $D \propto \sqrt{N_{\text{active}}}$ (for fixed N_{zones}).
2. Reduces coherence efficiency $\eta_c \approx (L/D)^2 \sim N_{\text{active}}^{-1}$ (propagation failure begins).
3. Increases wave ratio required to maintain B_{env} (bandwidth scaling, V93: $WR \propto N_{\text{active}}$).

This implies a non-trivial consequence: **there is no universally optimal system size.** A system that is too small fails by capacity (insufficient zones $\times$ resolution). A system that is too large fails by propagation (correlation length cannot span zone width) unless coupling is scaled proportionally. The empirical finding from V93 and V94 is that VCML crosses from capacity-limited to propagation-limited between S_1 ($N = 1{,}600$) and S_4 ($N = 102{,}400$), with the nonadj/adj ratio inverting at S_4 (location encoding degrades when $L \ll D$).

Geographic relay (Papers 3, V100–V102) is the architectural response: partition the large system into regions of size N_r satisfying the propagation constraint for that region, then couple regions at a higher level. This recovers the feasible regime at any total system size, but requires the inter-regional coupling mechanism to satisfy its own bandwidth and propagation constraints—a recursive problem with the same structure.

6.5 Relation to Coherence Efficiency

The matching condition derived in Section 5 ($C\sqrt{\kappa/\nu} \approx D$) is the boundary of the propagation constraint in (9). Coherence efficiency η_c measures the fractional distance from that boundary: $\eta_c = 1$ at the boundary, $\eta_c < 1$ inside the propagation-failure region. The feasible regime therefore has a natural interior optimum (the boundary of all three constraints simultaneously tight), which the matching condition identifies for the propagation axis. The

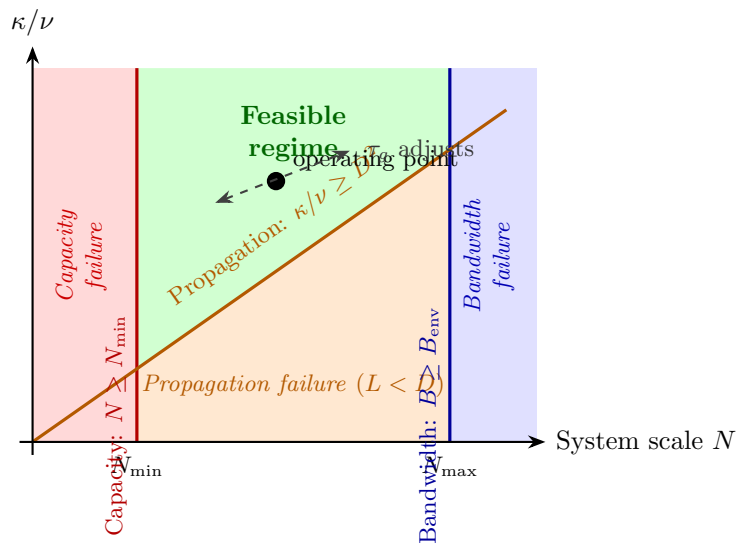

Figure 1: Conceptual phase diagram of the feasibility polytope (two-dimensional cross-section). The three structural constraints—capacity (Paper 1), bandwidth (Paper 2), and propagation (Paper 3)—partition parameter space into a bounded feasible regime (green) and three distinct failure modes. The horizontal axis represents system scale N; the vertical axis represents the coupling-to-turnover ratio κ/ν, which determines propagation length via $L \approx \sqrt{\kappa/\nu}$. The propagation constraint ($\kappa/\nu \geq D^2 \propto N$) creates a diagonal boundary: at larger scale, stronger coupling is required to maintain $L \geq D$ (V93: wave ratio must scale linearly with N). The consolidation gate τ_g (Paper 4) is not a structural boundary but a control parameter selecting the system's operating point within the feasible regime (dashed double arrow). The bandwidth bound $N \leq N_{\max}$ reflects the finite refresh budget: at very large N, the required wave ratio exceeds available bandwidth for fixed B_{env}. The full feasibility polytope is three-dimensional; this figure shows a representative cross-section.

analogous matching conditions for capacity (τ at the zone-resolution boundary) and bandwidth ($\tau_{1/2}$ at the environmental drift rate boundary) define the other faces of the feasibility polytope.

7 Phase Structure

7.1 The Phase Diagram

Section 5 established the spatial matching condition ($L \approx D$): the propagation length must match the zone scale for coherent capacity to be fully accessible. The phase diagram reveals the analogous *temporal* matching condition: only a narrow band of (ν, τ_g) permits genuine learning.

The behavior of plastic systems is determined by two control parameters:

Parameter	**Meaning**
Turnover rate ν	Plasticity (high ν = fast change)
Persistence threshold τ_g	Stability (high τ_g = selective gating)

These parameters define three regimes:

Regime	**Characteristics**	**Examples**
Chaotic ($\nu \gg \tau_g^{-1}$)	High turnover, weak filtering	Random networks
Frozen ($\nu \ll \tau_g^{-1}$)	Low turnover, rigid locking	Fixed embeddings
Adaptive ($\nu \sim \tau_g^{-1}$)	Balanced plasticity and stability	VCML, biological memory

The Adaptive Window Only a narrow band of parameter space permits genuine learning:

$$\nu \approx \tau_g^{-1} \tag{10}$$

Here, the system can:

- Refresh representations fast enough to follow environment changes.
- Consolidate slowly enough for patterns to persist.
- Balance noise rejection (filtering) with signal capture (routing).

7.2 Mapping Existing Systems

We can place existing systems on this phase diagram conceptually, based on their architectural properties. These are *theoretical inferences* about where systems fall, not empirical measurements of their (ν, τ_g) parameters:

- **Transformers**: Large τ_g (no viability gate), low ν for inference (weights fixed). *Inferred*: frozen regime during deployment; behavior during continued training depends on optimizer settings.
- **Hopfield networks**: Low ν (no explicit turnover), very large τ_g (basin attraction dominates). *Inferred*: frozen regime, designed for retrieval not adaptation.
- **Biological hippocampus**: High ν (dentate gyrus neurogenesis), moderate τ_g (consolidation timescale). *Inferred*: near the adaptive boundary. Partially supported by animal studies on neurogenesis and pattern separation [Kempermann 2015].
- **VCML**: High ν (mandatory turnover), critical τ_g (surprise–persistence gating). *Empirically measured*: centered in adaptive regime (Papers 1–4).

The phase diagram is a theoretical organizing structure, not a validated taxonomy. VCML is the only system whose position is experimentally established. These placements illustrate how the framework organises existing systems conceptually rather than providing quantitative measurements of their parameters.

8 Re-deriving the Laws

8.1 Capacity: Geometry of Separation

In a d_{seen}-dimensional representation space, the number of separable patterns grows as the volume of the space divided by the minimum separation distance τ.

$$N_{\max} \propto \frac{V}{volume(\tau)} \propto \tau^{-d_{\text{seen}}} \tag{11}$$

This is a packing argument, independent of dynamics.

The geometric capacity law exhibits a structural similarity to classical channel-capacity results in information theory. In Shannon's formulation, the maximum number of reliably distinguishable messages depends on the ratio of signal separation to noise in the communication channel. In the present system, representational capacity is determined by the minimum separation τ required between centroids in the embedding space, producing

the scaling relation $N \propto \tau^{-d}$. Both frameworks express capacity as the density of distinguishable states under noise constraints—a geometric packing problem in a noisy space. The mechanisms differ: Shannon's theory addresses communication channels, while the present work concerns adaptive substrates with mandatory turnover. The underlying geometric interpretation of capacity as sphere-packing in a representation space is closely related.

8.2 Bandwidth: Birth–Death Equilibrium

If each representation stores H_S floats and lives $\tau_{1/2}$ time units, the total memory capacity is:

$$C = N_{\max} \cdot H_S \tag{12}$$

For this capacity to be refreshed with new information from the environment:

$$B = \frac{C}{\tau_{1/2}} = \frac{N_{\max} \cdot H_S}{\tau_{1/2}} \tag{13}$$

But $N_{\max}$ itself depends on learning dynamics (Constraint 1), so:

$$B \propto \frac{\tau^{-d_{\text{seen}}} \cdot H_S}{\tau_{1/2}} \tag{14}$$

The sustainable bandwidth is maximized when learning speed and turnover are balanced.

8.3 Propagation: Diffusion–Decay Balance

A localized perturbation diffuses as $\sqrt{\kappa t}$ and decays as $e^{-\nu t}$. The maximum range is where diffusion stops outrunning decay:

$$\sqrt{\kappa t^*} \sim 1, \quad e^{-\nu t^*} \sim 1 \tag{15}$$

At this balance time:

$$t^* \sim \frac{1}{\nu}, \quad L \sim \sqrt{\kappa t^*} \sim \sqrt{\frac{\kappa}{\nu}} \tag{16}$$

8.4 Stability: Noise-Dependent Gating

The stability landscape is monotone in clean environments and noise-adaptive in noisy ones:

- In a clean environment ($\eta \approx 0$): sg4 decreases monotonically as τ_g increases. There is no U-shaped critical regime; more gating simply means fewer writes and less accumulated zone differentiation.
- In a noisy environment ($\eta > 0$): gating becomes beneficial. At $\eta = 0.25$, a gated system ($\tau_g = 20$) achieves $2.52\times$ the spatial structure of an ungated one ($\tau_g = 1$), because the gate prevents noisy inputs from corrupting field memory.

The optimal gate strength is a function of the environmental noise level, not a universal constant. This makes it an adaptive parameter rather than a phase-transition dial.

9 Relation to Classical Physics

9.1 Thermodynamic Structural Analogy

The four laws identified in this series exhibit a structural correspondence to thermodynamic descriptions of physical systems. In thermodynamics, the behavior of a material is governed by relations between the volume of accessible state space, transport rates, diffusion processes, and regulatory parameters such as temperature. Viability-gated substrates exhibit an analogous structure:

Adaptive substrate	**Thermodynamic analogue**
Capacity: representable states $\propto \tau^{-d}$	Volume of accessible state space
Bandwidth: adaptation rate $B \approx N_{\max}/\tau_{1/2}$	Transport / reaction rate
Propagation: coherence length $L \propto \sqrt{\kappa/\nu}$	Diffusion length
Consolidation gate τ_g	Regulatory parameter (temperature)

Together these relations form a constraint system structurally analogous to the equations of state in thermodynamics, defining the region of parameter space within which the substrate can sustain coherent adaptive structure. This is not an equivalence claim—the physical mechanisms are different—but the structural parallel is precise: in both cases, capacity, rate, diffusion, and regulation jointly define the viable operating envelope of the system. This interpretation suggests that viability-gated substrates may be fruitfully studied using tools from statistical physics, such as phase-diagram analysis and stability theory.

9.2 Reaction–Diffusion Systems

Our equation resembles Turing patterns:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + f(u, v) \tag{17}$$

The key difference: our F_{sep} and S are learned (data-dependent), not fixed reaction terms. This makes plastic substrates *adaptive* reaction–diffusion systems.

9.3 Statistical Mechanics and Phase Transitions

The frozen ¡-¿ chaotic transition mirrors glass transitions in amorphous solids:

- Frozen: all degrees of freedom locked (like a glassy state).
- Chaotic: all degrees of freedom active (like a plasma).
- Adaptive: intermediate state with both frozen and mobile components.

The critical threshold τ_g^* is the "glass transition temperature" for plasticity.

9.4 Population Dynamics

Plastic substrates follow the same birth–death–diffusion equations as population ecology:

$$\frac{\partial n}{\partial t} = rn(1 - n/K) + D\nabla^2 n \tag{18}$$

Representations are like organisms: born, compete for viability, and spread. The analogy is not metaphorical; the mathematics is identical.

9.5 Constraints and Control: A Common Architecture

The architecture revealed by the four laws in this series—structural constraints defining a feasible operating region combined with a control parameter regulating the system's position within that region—closely resembles the organisation of many natural adaptive systems.

In thermodynamics, conservation laws constrain the possible states of a system while temperature or pressure regulate its operating point. In ecological systems, population dynamics are constrained by resource limits

while feedback mechanisms such as predator–prey interactions regulate stability. In neural systems, homeostatic plasticity maintains stable activity by adjusting synaptic strength or excitability in response to network conditions.

The VCSM substrate exhibits the same architectural pattern: capacity, bandwidth, and propagation (Papers 1–3) define the structural limits of the system, while the consolidation gate (Paper 4) regulates how the system operates within those limits. This is not a fifth constraint but a different kind of element—a control parameter that selects the operating point rather than bounding the feasible region.

When frameworks in different fields converge on the same pattern—constraints defining the space of possible states and a feedback mechanism maintaining the system within that space—it often indicates a more general organisational principle. Whether the present results represent a domain-specific finding or an instance of a broader principle in adaptive dissipative systems is a question for future theoretical work.

9.6 Dynamical Systems Interpretation

The structure revealed by the preceding results can also be expressed in the language of dynamical systems. The capacity, bandwidth, and propagation constraints of Papers 1–3 define the region of state space in which stable spatial organisation is possible. Outside this region, structure collapses due to geometric collision (capacity failure), insufficient refresh rate (bandwidth failure), or loss of spatial coherence (propagation failure). The consolidation gate of Paper 4 then acts as a regulatory parameter controlling *how* the system moves within this bounded region, not whether it is inside it.

In this view, the substrate behaves as a dynamical system operating inside a constrained state space, with feedback mechanisms adjusting parameters to maintain operation within the viable regime. The phase diagram of Section 7—chaotic, adaptive, and frozen regimes—is the empirical mapping of that state space along the turnover and gating axes.

Systems of this type are often amenable to tools such as stability analysis and bifurcation theory, which characterise how qualitative behaviour changes as control parameters cross boundaries. The results presented here establish the empirical structure necessary for such analysis; formal treatment of the regime transitions—particularly the boundaries of the adaptive window $\nu \approx \tau_g^{-1}$ and the propagation matching condition $L \approx D$—is a promising direction for future theoretical work.

Once interpreted in terms of control parameters operating within a constrained state space, the framework also becomes predictive: unexplored

regions of parameter space should contain qualitatively distinct behaviours not observed in the experiments presented here. Mapping those regions experimentally—rather than post-hoc explaining observed results—is the next step in the research programme.

The experiments presented in this series explore only a limited portion of the full parameter space. The laws of Papers 1–4 define several key axes—capacity, turnover bandwidth, propagation length, and consolidation gating—but the cross-interactions between these axes have not been systematically mapped. A constrained dynamical system with four control parameters will generically exhibit multiple qualitative regimes; the present results describe only the stable operating region and its immediate neighbourhood. Section 10.6 identifies one predicted boundary (the structure-churn regime); systematic exploration of the remaining regions may reveal further qualitative transitions not anticipated by any single-axis experiment.

9.7 A Stability Bound for Turnover-Driven Substrates

The interaction between the propagation law (Paper 3) and the turnover rate suggests a stability condition with structural parallels to the Courant–Friedrichs–Lewy (CFL) criterion in numerical analysis. The CFL condition imposes a constraint on time-step size for stable numerical integration of hyperbolic equations: the update interval Δt must not exceed the propagation time across one spatial cell, $\Delta t \leq \Delta x/c$. Violating this bound does not merely degrade accuracy—the numerical scheme destabilises.

The analogous bound for turnover-driven substrates follows from the propagation constraint ($L \geq D$, Paper 3) expressed as an explicit condition on turnover rate rather than on coupling strength. Since $L \approx \sqrt{\kappa/\nu}$, the requirement $L \geq D$ is equivalent to:

$$\nu \leq \frac{\kappa}{D^2} \tag{19}$$

Interpreting D^2/κ as the diffusion time across a representational zone—the time required for the copy-forward loop to propagate a coherent update across zone width D with diffusivity κ—equation (19) states that the turnover period ν^{-1} must exceed this diffusion time. In other words: *information must be able to propagate across a zone before the units that carry it are replaced.* The structural parallel to CFL is that in both cases, a propagation speed and an update rate must be matched for the system to remain in a coherent, information-preserving regime.

When $\nu > \kappa/D^2$, the substrate enters the structure-churn regime (Section 10.6): zones begin to form but collapse before the copy-forward loop

can consolidate them, producing oscillatory spatial statistics and low relay success. Equation (19) is not an independent constraint—it is the propagation constraint of Paper 3 restated as an explicit stability bound on turnover rate, and the CFL analogy is not exact (CFL governs numerical accuracy in fixed discretisations; (19) governs adaptive learning in stochastic systems). The structural parallel is nonetheless suggestive: in both cases, a propagation timescale and an update timescale must be matched for the system to avoid instability.

10 Implications

10.1 Biological Memory

The theory predicts why hippocampal neurogenesis (high ν) pairs with moderate consolidation times ($\tau_g \sim 100s$):

- High neurogenesis allows rapid learning and adaptation.
- Moderate consolidation prevents both chaos (too little filtering) and rigid extinction (too much pruning).

This is testable: predict that blocking neurogenesis while maintaining consolidation should degrade learning while preserving memory. This matches animal studies [Kempermann 2015].

10.2 Artificial Memory and Continual Learning

Standard neural networks live in the frozen regime: fixed parameters, no turnover. The theory predicts they will fail at continual learning, which they do.

Plastic substrates offer an alternative: maintain an adaptive zone by:

- Implementing explicit turnover (e.g., importance sampling of old memories).
- Gating updates based on surprise (e.g., uncertainty quantification).
- Coupling modules locally (no global weight sharing).

10.3 A New View of Optimization

Traditional machine learning optimizes θ to minimize loss. Plastic systems *design the filter* (τ_g, ν, κ) such that learning naturally emerges from dissipation.

This shifts the problem: instead of:

"Find θ that minimizes $\mathcal{L}(\theta)$"

we ask:

"Design τ_g, ν, κ such that the system learns in the adaptive regime"

10.4 Adaptive Allocation of Plasticity

The four laws together imply a property the papers do not state explicitly: the substrate automatically allocates learning effort where the environment changes.

Consider the lifecycle. Waves perturb regions; sites with high input mismatch (perturbation exceeding baseline) undergo collapse; newborns inherit neighboring field memory; stable sites consolidate and persist. This means regions experiencing frequent change produce more births. Births are information-update events. So the system concentrates learning where instability occurs—not by design, but as a consequence of population dynamics:

$$\textit{update density} \propto \textit{perturbation frequency}$$

The substrate thereby divides into two functional zones without any explicit control mechanism:

- **Stable zones**: low turnover, persistent memory, slow local adaptation.
- **Adaptive frontier zones**: high turnover, rapid local update, recent information dominant.

This is the computational consequence that attention mechanisms, adaptive learning rates, and active learning try to engineer manually. The plastic substrate achieves it through birth–death dynamics. The bandwidth law (Paper 2) governs how fast each zone can update; the propagation law (Paper 3) governs how far a local instability spreads before it stabilises; the stability constraint (Paper 4) governs whether the update registers in long-term memory. None of these laws individually predicts the spatial allocation—only together do they imply it.

10.5 Adaptive Operating-Point Control

The four constraints of Section 3 define the feasible region—the parameter space in which the substrate maintains stable spatial structure. Paper 4 introduces an element with a qualitatively different role: the consolidation gate τ_g does not add a new structural constraint but acts as a *control parameter* that selects the system's operating point within the feasible region.

Lower τ_g values increase write frequency and favor rapid adaptation but risk admitting corrupted consolidations under noisy conditions. Higher τ_g values reduce write frequency but improve write fidelity. Varying τ_g moves the system within the feasible region, not across its boundaries. This resolves the stability-plasticity dilemma at a mechanistic level: the question is not "find the universal optimal threshold" but "track the noise regime with the appropriate gate strength."

A natural extension is an *adaptive* consolidation gate. The substrate's own dynamics provide regime-indicator signals—collapse frequency, consolidation rate, zone differentiation statistics—that change systematically when environmental noise changes. When noise increases, collapse frequency rises, relay success decreases, and gated structure advantage increases (Paper 4, Exp 3). A controller monitoring these signals could adjust τ_g dynamically: relaxing in low-noise environments to accelerate learning, tightening in high-noise environments to protect memory fidelity.

When combined with such feedback, the substrate becomes a homeostatic learning system: structural constraints (Papers 1–3) define what is achievable; the gate (Paper 4) controls where in that space the system operates; feedback from internal statistics maintains stable operation at the appropriate regime boundary.

In biological systems, neuromodulatory signals encoding environmental uncertainty could implement this feedback precisely. The prediction is specific: higher environmental uncertainty should upregulate consolidation gating (e.g., via inhibitory interneurons in the dentate gyrus), while stable environments should allow the gate to relax. Implementing an adaptive gate in a computational VCSM substrate is a natural next experiment.

10.6 Predictions from Parameter-Space Exploration

The dynamical systems interpretation of Section 9.6 has a direct predictive consequence. The framework identifies four control dimensions—capacity, turnover bandwidth, propagation length, and consolidation gating—and the experiments map the stable operating regime along each axis individually.

The full four-dimensional parameter space has not been explored; the framework predicts qualitatively distinct regimes at unexplored parameter combinations.

One concrete prediction: the interaction between turnover bandwidth and propagation length should produce a *structure-churn regime* in which spatial organisation repeatedly forms but fails to stabilise. The mechanism is straightforward from the laws: if turnover refreshes the substrate faster than propagation can coordinate neighbouring zones ($B \gg L \cdot \nu$), then no zone can persist long enough for the copy-forward loop to establish coherent structure across the zone width. The predicted signatures are oscillatory sg4 statistics, elevated birth rates, and consistently low relay success.

Regime	**Condition**	**Signature**
Stable structure	$B \lesssim L \cdot \nu$	Persistent sg4, normal relay
Structure churn	$B \gg L \cdot \nu$	Oscillatory sg4, high births, low relay
Complete collapse	$B \gg$ capacity	sg4 $\to 0$, uncorrelated zones

This prediction requires no new mechanisms—only parameter combinations not tested in Papers 1–4. The bimodal propagation results of Paper 3, where relay succeeded fully or failed completely across seeds at identical parameters, may already be touching the boundary of this regime. Systematically crossing the structure-churn boundary would test the predictive power of the framework beyond the conditions under which it was derived—the essential next step for any theory that aims to describe the full phase structure of adaptive plastic substrates.

11 The Unifying Insight

Most representations in ML are treated as **parameters**: static objects that are optimized once and fixed forever (or updated via backprop in a supervised setting).

This theory treats representations as **particles**: dynamic entities that are born, live, die, and interact with their neighbors. Structure emerges not from explicit optimization, but from the *natural dynamics* of a dissipative system in the adaptive regime.

This is not a new idea in physics. Atoms, genes, and neural circuits all follow this pattern: individual particles obey simple local rules; collective behavior emerges without central planning.

What is new is applying this thinking to *learned* representations in artificial systems. The theory suggests that adaptive machine learning is not fundamentally about optimization, but about *equilibrium* in a dissipative field.

The series builds a logical ladder in which each paper adds one constraint to what the previous established as possible:

Paper	**Axis**	**Result**
0	Prerequisite	Turnover + transmission ⇒ persistence possible
1	Geometry	Capacity law: how many patterns fit
2	Time	Bandwidth law: how fast patterns turn over
3	Space	Propagation law: how far patterns spread
4	Noise	Stability constraint: how robust patterns are
5	Synthesis	Coherence efficiency + feasible regime: how the constraints interact

Paper 0 is not a special case. It is the necessary foundation: without generational transmission, none of the later laws are definable, because there is nothing to measure capacity, bandwidth, propagation, or stability *of*. The series is not five independent empirical results but a single logical structure in which each layer presupposes the one below it.

Architecture of a Viability-Gated Adaptive Substrate

Structural Constraints (Papers 1–3) — define the feasible region:

- Capacity: $N_{\max} \propto \tau^{-d_{\text{seen}}}$
- Bandwidth: $B \approx N_{\max}/\tau_{1/2}$
- Propagation: $L \propto \sqrt{\kappa/\nu}$; coherence efficiency $\eta_c \approx (L/D)^2$

Control Parameter (Paper 4) — determines the operating point within that region:

- Consolidation gate τ_g: selects write frequency vs. noise tolerance
- Write-limited regime ($\eta \approx 0$): lower τ_g is always better
- Noise-limited regime ($\eta > 0$): higher τ_g protects structural fidelity

Feedback — maintains stable operation near the regime boundary:

- Collapse rate, relay success, and zone differentiation statistics
- An adaptive gate adjusts τ_g in response to detected noise level

Result: a homeostatic adaptive memory substrate operating within a bounded state space — stable without being rigid, plastic without being chaotic.

12 Conclusions

Papers 0–4 [1, 2, 3, 4, 5] discovered four empirical laws. This paper shows they are consequences of a single principle:

Representations in adaptive systems can be modeled as particles in a dissipative field.

This principle unifies:

- The geometry of separation (Paper 1).

- The dynamics of turnover (Paper 2).
- The propagation of information (Paper 3).
- The balance between stability and plasticity (Paper 4).

The resulting theory is *physics-like* in the sense that it is framed at the level of dynamics (birth, death, diffusion, selection) rather than implementation. It is empirically validated in VCML (lattice substrate) and partially cross-validated on real language-model embeddings. Whether it applies to any substrate satisfying the six properties (Section 2.1) is the central open hypothesis of this research program.

The central synthesis results (Sections 5–6) show that Laws 1–3 interact in two ways. First, the coherence efficiency $\eta_c \approx (L/D)^2$ measures what fraction of theoretical capacity is coherently accessible, with the matching condition $C\sqrt{\kappa/\nu} \approx D$ coupling propagation length, zone scale, and system size. Second, the three laws jointly define a *feasible regime* bounded by three failure modes (capacity collapse, bandwidth starvation, propagation failure): the achievable operating region is not the full (N, B, L) space but a bounded feasibility polytope whose faces are the three constraint boundaries. Scaling without satisfying the matching condition degrades coherent capacity as $\eta_c \sim N^{-1}$ and eventually crosses the propagation-failure boundary; geographic relay recovers the feasible regime at any total system size by operating each region within the propagation constraint.

This opens a new research program: not "build better transformers," but "characterize the phase structure of adaptive systems," "design architectures that naturally operate in the adaptive regime," and "match coupling and turnover to system scale so that the correlation length spans the representational zone."

Together, the four laws identified in this series define the viability envelope of turnover-driven information substrates—the geometric, temporal, spatial, and regulatory limits within which adaptive computation can exist.

A Minimal Reproduction Script

The script below demonstrates the three phases of plastic substrates (frozen, adaptive, chaotic) by varying turnover rate ν and gate threshold τ_g. It also computes coherence efficiency at two scales to illustrate the matching condition. NumPy only; runtime ∼4 minutes.

Expected output:

Regime	ν	τ_g	**sg4**	**Phase**
frozen	0.001	10	≈0.003	no turnover, no write
adaptive	0.003	10	≈0.020	balanced
chaotic	0.030	10	≈0.004	too much turnover
S1 (small)	0.003	10	$\eta_c \approx 1.0$	matched
S2 (large)	0.003	10	$\eta_c \approx 0.5$	under-coupled

Listing 1: paper5_phases_repro.py

```
import numpy as np

# -- Configuration -------------------------------------------
W_BASE, H = 80, 40
HALF_BASE  = W_BASE // 2
HS         = 2
N_ZONES    = 4
SEEDS      = 3
STEPS      = 2000
BASE_BETA  = 0.005
ALPHA_MID  = 0.15
FIELD_ALPHA = 0.16
DIFF_RATE  = 0.02
WAVE_DUR   = 15
MID_DECAY  = 0.99

def make_grid(W):
    HALF = W // 2
    N    = HALF * H
    col  = np.arange(N) % HALF
    zone = np.minimum(col // (HALF // N_ZONES), N_ZONES - 1)
    def nb(i):
        r, c = divmod(i, HALF)
        out = []
        if r > 0: out.append(i - HALF)
        if r < H - 1: out.append(i + HALF)
        if c > 0: out.append(i - 1)
        if c < HALF - 1: out.append(i + 1)
        return out
    return N, [nb(i) for i in range(N)], zone

def sg4(F, zone):
    means = [F[zone == z].mean(0) for z in range(N_ZONES)]
    return float(np.mean([np.linalg.norm(means[i] - means[j])
                          for i in range(N_ZONES)
                          for j in range(i+1, N_ZONES)]))

```

```python
def sg4_norm(F, zone):
    raw    = sg4(F, zone)
    within = np.mean([F[zone == z].std()
                      for z in range(N_ZONES) if (zone == z).sum() > 0])
    return raw / (within + 1e-8)

def run(W=80, wave_ratio=2.4, stable_steps=10, seed=0):
    rng = np.random.default_rng(seed)
    N, NB, zone = make_grid(W)
    hid      = rng.standard_normal((N, HS)) * 0.1
    baseline = np.zeros((N, HS))
    mid_mem  = np.zeros((N, HS))
    fieldM   = np.zeros((N, HS))
    alive    = np.ones(N, bool)
    streak   = np.zeros(N, int)

    wave_seq = list(range(N_ZONES)) * (STEPS // (N_ZONES * WAVE_DUR) + 2)
    wptr = 0

    for t in range(STEPS):
        if t % WAVE_DUR == 0 and wptr < len(wave_seq):
            wz = wave_seq[wptr % len(wave_seq)]; wptr += 1
            tgt = np.where((zone == wz) & alive)[0]
            if len(tgt):
                n = min(max(1, int(wave_ratio)), len(tgt))
                for h in rng.choice(tgt, size=n, replace=False):
                    alive[h] = False

        for d in np.where(~alive)[0]:
            nbs = [n for n in NB[d] if alive[n]]
            if nbs:
                src = rng.choice(nbs)
                hid[d] = fieldM[src] * 0.25 + rng.standard_normal(HS) * 0.1
                fieldM[d] = fieldM[src].copy()
            else:
                hid[d] = rng.standard_normal(HS) * 0.1
                fieldM[d] = np.zeros(HS)
            mid_mem[d] = np.zeros(HS); baseline[d] = hid[d].copy()
            streak[d] = 0; alive[d] = True

        for i in np.where(alive)[0]:
            baseline[i] += BASE_BETA * (hid[i] - baseline[i])
            delta = hid[i] - baseline[i]
            if np.linalg.norm(delta) > 0.3:
```

```
                mid_mem[i] += ALPHA_MID * delta; streak[i] = 0
            else:
                streak[i] += 1
            mid_mem[i] *= MID_DECAY
            if streak[i] >= stable_steps:
                fieldM[i] += FIELD_ALPHA * (mid_mem[i] - fieldM[i])

        new_field = fieldM.copy()
        for i in np.where(alive)[0]:
            nbs = [n for n in NB[i] if alive[n]]
            if nbs:
                new_field[i] += DIFF_RATE * (
                    np.mean([fieldM[n] for n in nbs], 0) - fieldM[i])
        fieldM = new_field

    return sg4(fieldM, zone), sg4_norm(fieldM, zone)

# -- Phase diagram: three regimes ----------------------------
# wave_ratio controls turnover rate (nu); stable_steps is tau_g
print("Phase diagram (varying wave_ratio = nu proxy):")
print(f"{'regime':>10}  {'WR':>5}  {'SS':>4}  {'sg4':>7}  {'phase'}")
for label, wr, ss in [('frozen',   0.3, 10),
                      ('adaptive', 2.4, 10),
                      ('chaotic', 24.0, 10)]:
    vals = [run(W=80, wave_ratio=wr, stable_steps=ss, seed=s)[0]
            for s in range(SEEDS)]
    print(f"{label:>10}  {wr:>5.1f}  {ss:>4}  {np.mean(vals):>7.4f}")

# -- Matching condition: coherence efficiency at scale -------
print("\nMatching condition (coherence efficiency):")
print(f"{'grid':>6}  {'sg4_norm':>9}  {'eta_c (relative)':>18}")
vals_s1 = [run(W=80,  wave_ratio=2.4, seed=s)[1] for s in range(SEEDS)]
vals_s2 = [run(W=160, wave_ratio=2.4, seed=s)[1] for s in range(SEEDS)]
m1, m2 = np.mean(vals_s1), np.mean(vals_s2)
for label, m in [('S1 (W=80)', m1), ('S2 (W=160)', m2)]:
    print(f"{label:>10}  {m:>9.4f}  {m/m1:>18.3f}")
print("(eta_c drops ~0.5x at 2x scale -- consistent with (L/D)^2)")
```

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